

Arista CVaaS AQL Notebook — All Transceiver Inventory to CSV

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Overview

This guide walks through generating a full transceiver inventory (hostname, device ID, interface, transceiver type, and serial number) for every switch in a CVaaS tenant, using the AQL notebook plus a local merge script.

Related files in this process:

- “[Transceiver-Inventory-AQL-Notebook.txt](#)” — the three AQL queries to paste into the notebook
- “[merge_transceivers.py](#)” — combines the three CSV exports into the final report

1. Log into CVaaS

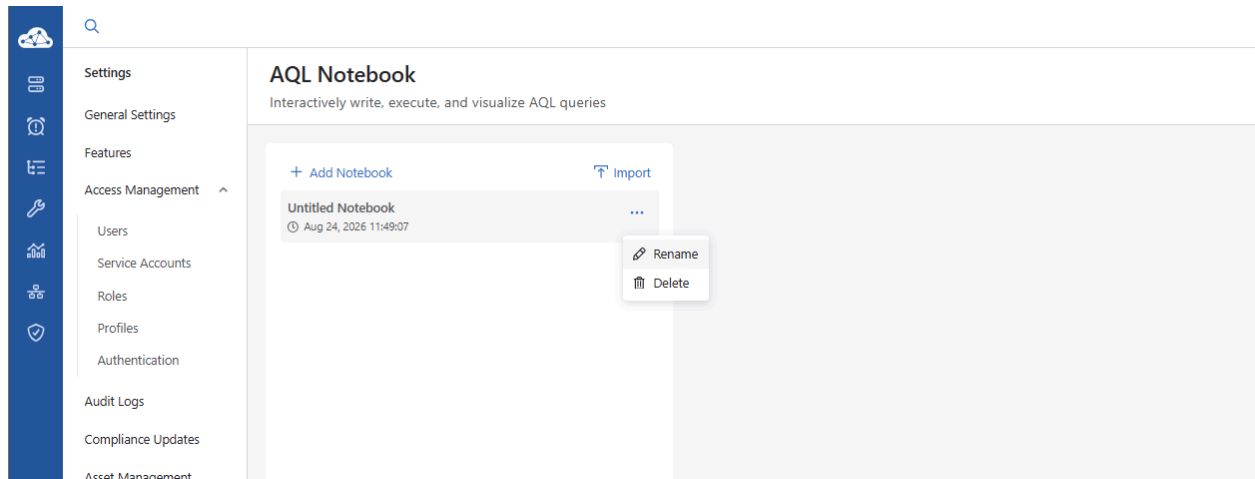
Log into your CloudVision as-a-Service tenant in a browser (e.g. `https://www.cv-prod-us-central1-c.arista.io``, or whichever cluster hosts your tenant).

2. Navigate to the AQL Notebook

1. Click the “**Settings**” (the gear icon in the lower left panel).
2. Go to **Settings** → **Developer Tools** → **AQL Notebook**

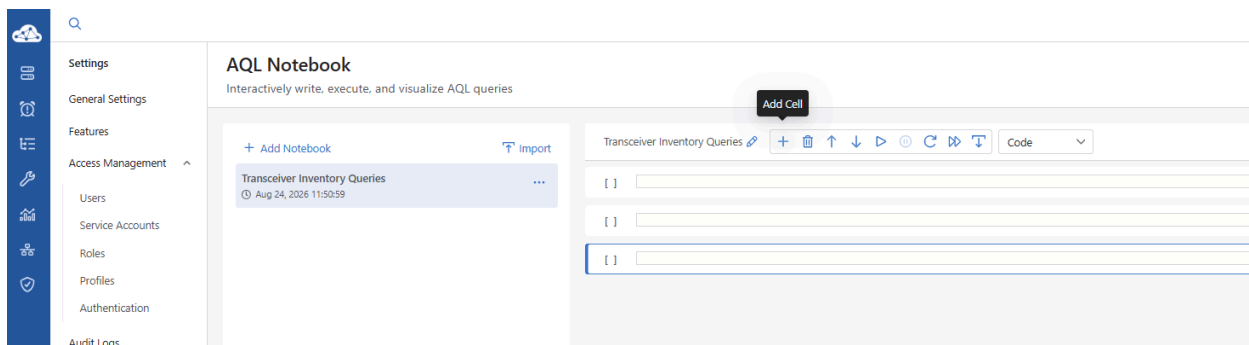
3. Create the notebook

1. Click the “**+ Add Notebook**” button at the Top.
2. Rename the notebook using the three dots to the left. Example “Transceiver Inventory Queries”



4. Add and run the three query cells

1. Open `Transceiver-Inventory-AQL-Notebook.txt` file for the exact query text. Add three cells to the notebook, one per query:
 - a. To create a new Cell, click on the “+” symbol at the top of the Cells field. Create a total of 3 Cells



2. Paste the “Cell 1 — Serial number, per device/interface” query text from Transceiver-Inventory-AQL-Notebook.txt into the first Cell
3. Paste the “Cell 2 — Media/form-factor type, per device/interface” query text from Transceiver-Inventory-AQL-Notebook.txt into the second Cell
4. Paste the “Cell 3 — Hostname, per device” query text from Transceiver-Inventory-AQL-Notebook.txt into the third Cell

5. Run all three cells by clicking the “Run All” button (two play buttons).

The screenshot shows the AQL Notebook interface. On the left is a sidebar with navigation options like Settings, General Settings, Features, Access Management, Users, Service Accounts, Roles, Profiles, Authentication, Audit Logs, Compliance Updates, Asset Management, and Licenses. The main area is titled 'AQL Notebook' and contains three code cells under the heading 'Transceiver Inventory Queries'. A 'Run All' button is visible at the top right of the code area.

6. Export the result to CSV using the notebook's export option, and save it with the filename noted below (all three files should end up in the same local folder, alongside `merge\_transceivers.py`, to keep the next step simple).

a. After all three Cells have run, select the “Table” option for each cell on the right hand side.

The screenshot shows the AQL Notebook interface with the results of a query displayed in a table format. The table has four columns: 'key', 'Ethernet1', 'Ethernet1/1', 'Ethernet1/2', and 'Ethernet1/3'. The first two rows show data for 'WTW23230296' and 'WTW24220003'. A 'Download Button' and a 'Table Button' are visible on the right side of the table.

key	Ethernet1	Ethernet1/1	Ethernet1/2	Ethernet1/3
WTW23230296	—	—	—	—
WTW24220003	—	XHN213809375	XDP2016A0150	Unknown

b. Click on the Download button and save the file per the cell names given below.

Cell	Purpose	Export File Name
1	Transceiver serial number, per device/interface	serials.csv

2	Transceiver media/form-factor type, per device/interface	types.csv
3	Hostname, per device	hostnames.csv

5. Run the merge script

1. With Python 3 installed, open a terminal in the folder containing the three CSV files and `merge\_transceivers.py`, then run:

```
python merge_transceivers.py serials.csv types.csv hostnames.csv -o transceiver-report.csv
```

2. This produces `transceiver-report.csv`, the final long-format inventory.

6. What's in the final CSV

`transceiver-report.csv` has one row per installed transceiver — ports with no transceiver plugged in are automatically excluded, so this is a list of optics actually in use, not every physical port.

hostname	device_id	interface	type	serial
Goss-710	WTW23230296	Ethernet17	xcvr1000BaseT	XHT213360080
Goss-710	WTW23230296	Ethernet18	xcvr25GBaseC rN	XDY222802JR 8
ZTX-7250	WTW24220003	Ethernet1/1	xcvr10GBaseLr	XHN21380937 5

ZTX-7250	WTW24220003	Ethernet1/2	xcvr10GBaseLr	XDP2016A0150
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Description of Each Field

- **Hostname** - The device's configured hostname (e.g. `Goss-710`). Falls back to the device ID if no hostname was found.
- **Device_id** - The device's CVaaS device ID / serial number (e.g. `WTW23230296`).
- **Interface** - The interface the transceiver is installed in (e.g. `Ethernet17`).
- **Type** - The transceiver's media/form-factor type as reported by EOS (e.g. `xcvr1000BaseT`, `xcvr25GBaseCrN`).
- **Serial** - The transceiver's own vendor serial number (e.g. `XHT213360080`), distinct from the switch's device ID/serial above

7. Re-running later

To refresh the inventory, re-run steps 4–5: re-run and re-export the three notebook cells (they always reflect current fleet state), then re-run the merge script over the new exports.